PARISHRAM 2.0

CLASS 12TH 2026

d & f-Block

CHEMISTRY | LECTURE-6

BY SANYA NAYAR MA'AM



TOPICS TO BE COVERED

1. Potassium dichromate ✓
2. Lanthanoids ✓




ROAR
Tiger hai tu Tiger! Roar

Let's Recap

When Tiger Ma'am says-

Pehle zara purana revise kar lete hai



Potassium Dichromate ($K_2Cr_2O_7$)

(orangish)

Preparation: It is prepared from chromite ore in the following steps:

i) Chromite ore is fused with sodium carbonate in the presence of air to give sodium chromate.

$$2FeCr_2O_4 + 4Na_2CO_3 + {}^7/_2\,O_2 \rightarrow Fe_2O_3 + 4Na_2CrO_4 + 4CO_2$$

Chromite ore — Sodium Carbonate — Sodium chromate

ii) Na_2CrO_4 is filtered and acidified with conc. H_2SO_4 to give $Na_2Cr_2O_7$.

$$2Na_2CrO_4 + 2H^+ \rightarrow Na_2Cr_2O_7 + 2Na^+ + H_2O$$

Sodium Chromate — Sodium Dichromate

iii) Sodium dichromate solution is treated with KCl to give $K_2Cr_2O_7$.

$$Na_2Cr_2O_7 + 2KCl \rightarrow K_2Cr_2O_7 + 2NaCl$$

Sodium Dichromate — Potassium Dichromate

Potassium Dichromate ($K_2Cr_2O_7$)

Properties:

a) It is an orange crystalline solid with M.Pt 670 K.

b) It is appreciably soluble in hot water but moderately soluble in cold water.

c) With alkali :

$$Cr_2O_7^{2-} + 2OH^- \rightarrow 2CrO_4^{2-} + H_2O$$

Dichromate ion — Chromate ion (yellow)

d) With acid :

$$2CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{2-} + H_2O$$

Dichromate ion (orange red)

Dichromate ion — Chromate ion

$Cr_2O_7^{2-} \underset{H^+}{\overset{OH^-}{\rightleftharpoons}} CrO_4^{2-}$

orange — yellow

Potassium Dichromate ($K_2Cr_2O_7$)

Properties:

e) On heating, it decomposes

$$4K_2Cr_2O_7 \rightarrow 4K_2CrO_4 + 2Cr_2O_3 + 3O_2$$

Potassium Dichromate → Potassium Chromate (Yellow) + green

f) It is a powerful oxidising agent.

↳ oxidant

In acidic Medium

Reduction Half Rxn

$$\overset{+6}{Cr_2O_7^{2-}} + 6e^- + 14H^+ \longrightarrow 2Cr^{3+} + 7H_2O$$

$$\text{Eq. wt.} = \frac{\text{Mol. wt.}}{n_f}$$

$$= \frac{294}{6}$$

$$= 49$$

$K_2Cr_2O_7$ oxidises

- $I^- \rightarrow I_2$
- $Fe^{2+} \rightarrow Fe^{3+}$
- $S^{2-} \rightarrow S$
- $SO_3^{2-} \rightarrow SO_4^{2-}$
- $S_2O_3^{2-} \rightarrow SO_4^{2-}$
- $Sn^{2+} \rightarrow Sn^{4+}$ (Stannus) (Stannic)

Potassium Dichromate ($K_2Cr_2O_7$)

Properties:

g) Chromyl chloride test:

When a mixture of metal chloride is heated with $K_2Cr_2O_7$ and conc. H_2SO_4, orange red fumes of chromyl chloride are formed.



$$NaCl + K_2Cr_2O_7 + H_2SO_4 \xrightarrow{\Delta} KHSO_4 + NaHSO_4 + H_2O + CrO_2Cl_2$$

H_2SO_4 (Conc)

$KHSO_4$: Pot. Hydrogen Sulphate

$NaHSO_4$: Sod. Hydrogen Sulphate

CrO_2Cl_2: Chromyl Chloride

The orange red vapours of Chromyl chloride are then passed into NaOH solution which turns yellow

$$CrO_2Cl_2 + NaOH \longrightarrow Na_2CrO_4 + NaCl + H_2O$$

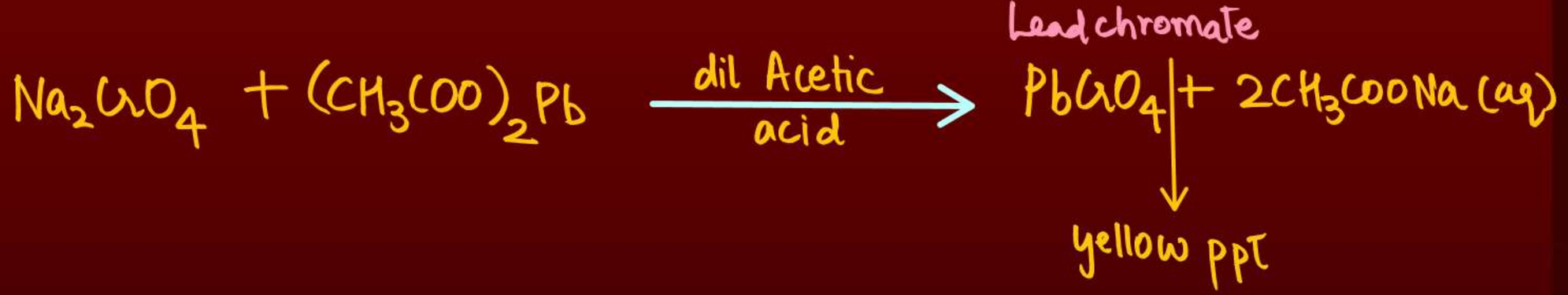
Na_2CrO_4: Sodium Chromate

Potassium Dichromate ($K_2Cr_2O_7$)

Properties:

g) Chromyl chloride test:

To this solution add lead acetate and a few drops of acetic acid



Potassium Dichromate ($K_2Cr_2O_7$)

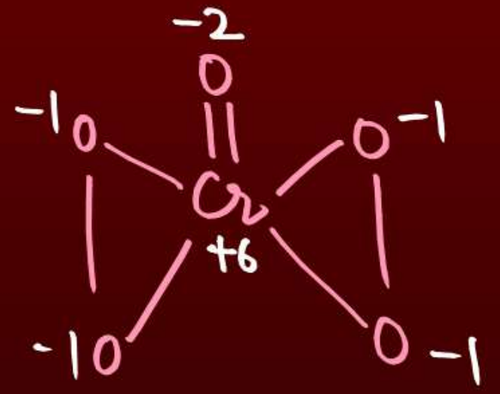
Properties:

h) Reaction with H_2O_2 :

$$Cr_2O_7^{2-} + 4H_2O_2 + 2H^+ \longrightarrow 2CrO_5 + 5H_2O$$

Dichromate ion

Hydrogen Peroxide



Potassium Dichromate ($K_2Cr_2O_7$)

Uses:

i) In leather industry for chrome tanning.

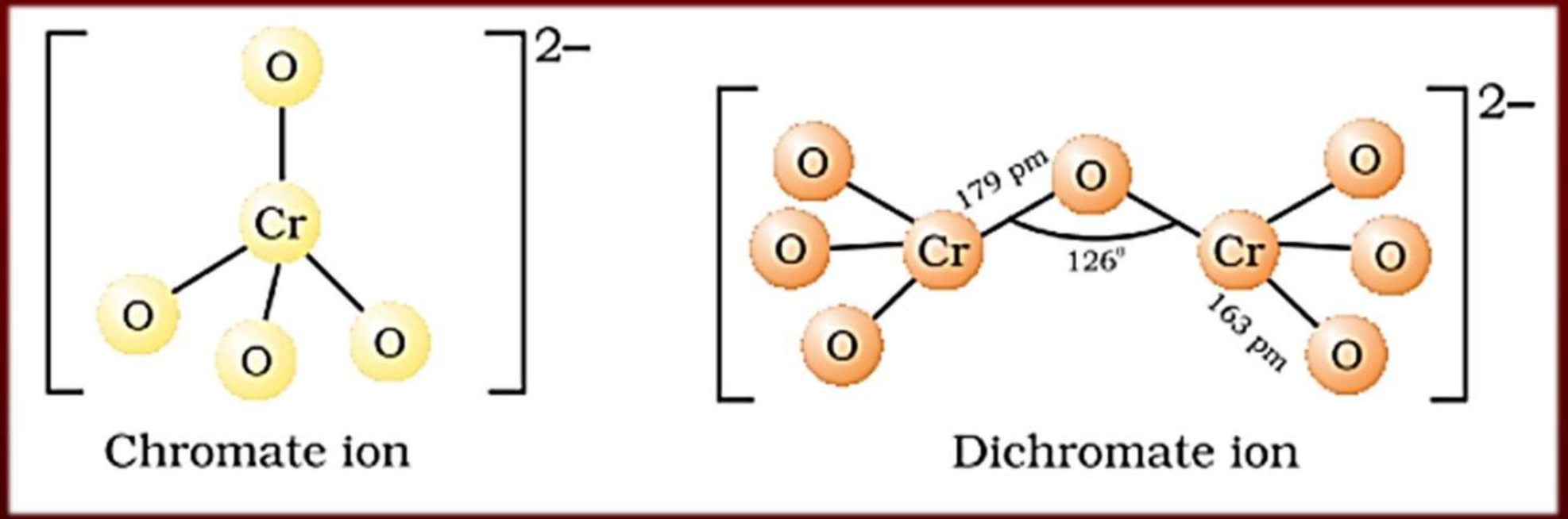
ii) For preparation of other chromium compounds such as chrome alum, chrome yellow etc.

iii) For the volumetric estimation of ferrous salts, iodides and sulphides.

iv) Used as an oxidizing agent.

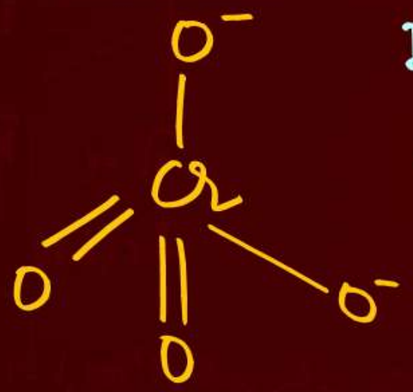
Potassium Dichromate ($K_2Cr_2O_7$)

Structure-



CrO_4^{2-}

Chromate ion: Yellow



Due to Resonance, all Cr-O bond lengths are equal

$Cr_2O_7^{2-}$

Dichromate ion: Orange



Due to Resonance, the 6 Cr-O bond lengths are equal.

QUESTION

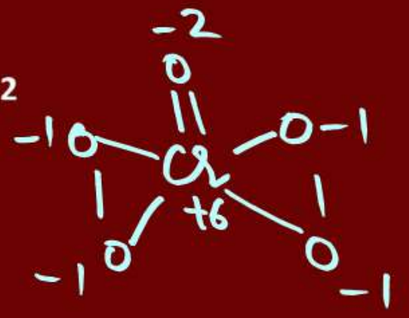
What is the oxidation no. of Cr in given compounds?

1. CrO_2Cl_2

$x - 4 - 2 = 0$

$x = +6$

2. CrO_5 or $CrO(O_2)_2$



3. $(NH_4)_2Cr_2O_7$

$Cr_2O_7^{2-}$

$x = +6$

QUESTION

Arrange the following ions in order of increasing strength as a reducing agent:

↳ oxidation (loss of e^-)

a) Cr^{2+}

b) Cr^{3+}

c) $Cr_2O_7^{2-}$ (+6)

$$Cr_2O_7^{2-} < Cr^{3+} < Cr^{2+}$$

The Inner Transition Elements (f-Block)

- The elements in which filling of electrons takes place in (n-2) f-subshell which belongs to anti-penultimate (third to the outermost) energy shell.

- The f-block consists of two series of elements known as lanthanoids and actinoids.

- The general electronic configuration of the f-block elements is

  $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$ For Lanthanoids, n is 6 while its value is 7 for actinoids.

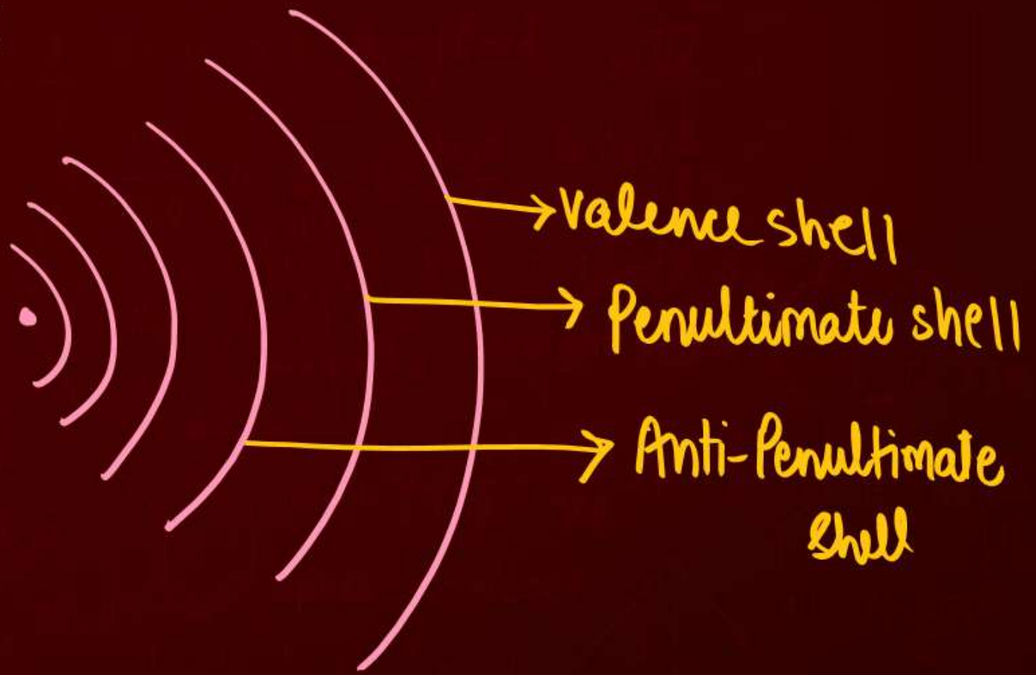
- There are many exceptions in the electronic configuration.

Lanthanoids [6th Period, $n=6$] → 4f

$(6-2) = (n-2)$ f-Subshell

Actinoids [7th Period, $n=7$] → 5f

$(7-2) = (n-2)$ f-Subshell



The Inner Transition Elements (f-Block)

2 series

f-Block

Lanthanoids

- $n=6$
- $\rightarrow$ $4f^{1-14}\,5d^{0-1}\,6s^2$
- $\rightarrow$ 6th Period
- $\rightarrow$ Filling of 4f
- $\rightarrow$ $6p^3$
- $\rightarrow$ $_{58}Ce - {}_{71}Lu$
- $\rightarrow$ Ist Inner Transition Series

Actinoids

- $n=7$
- $\rightarrow$ $5f^{1-14}\,6d^{0-1}\,7s^2$
- $\rightarrow$ 7th Period
- $\rightarrow$ Filling of 5f
- $\rightarrow$ $6p^3$
- $\rightarrow$ $_{90}Th - {}_{103}Lr$
- $\rightarrow$ 2nd Inner Transition Series

The Inner Transition Elements (f-Block)

Atomic Number	Name	Symbol	Electronic configurations*		
			Ln	Ln^{2+}	Ln^{3+}
57	Lanthanum	La	$5d^1 6s^2$	$5d^1$	$4f^0$
58 Celina	Cerium	Ce	$4f^1 5d^1 6s^2$	$4f^2$	$4f^1$
59 Preety	Praseodymium	Pr	$4f^3 6s^2$	$4f^3$	$4f^2$
60 Neha	Neodymium	Nd	$4f^4 6s^2$	$4f^4$	$4f^3$
61 Pammy	Promethium	Pm	$4f^5 6s^2$	$4f^5$	$4f^4$
62 Simmy	Samarium	Sm	$4f^6 6s^2$	$4f^6$	$4f^5$
63 Europe	Europium	Eu	$4f^7 6s^2$	$4f^7$	$4f^6$
64 Gayi	Gadolinium	Gd	$4f^7 5d^1 6s^2$	$4f^7 5d^1$	$4f^7$
65 Tab	Terbium	Tb	$4f^9 6s^2$	$4f^9$	$4f^8$
66 Diarrhoe	Dysprosium	Dy	$4f^{10} 6s^2$	$4f^{10}$	$4f^9$
67 Hug	Holmium	Ho	$4f^{11} 6s^2$	$4f^{11}$	$4f^{10}$
68 Erre	Erbium	Er	$4f^{12} 6s^2$	$4f^{12}$	$4f^{11}$
69 Tamaka	Thulium	Tm	$4f^{13} 6s^2$	$4f^{13}$	$4f^{12}$
70 Ye	Ytterbium	Yb	$4f^{14} 6s^2$	$4f^{14}$	$4f^{13}$
71 Liye	Lutetium	Lu	$4f^{14} 5d^1 6s^2$	$4f^{14} 5d^1$	$4f^{14}$

The series involving the filling of 4f-orbitals following Lanthanum (Z = 57) is called the lanthanoid series.

There are 14 elements in this series starting with Ce (Z=58) to Lu (Z=71)

General Electronic Configuration:

$[Xe]\ 4f^{1-14}\ 5d^{0-1}\ 6s^2$

Atomic and ionic sizes:

- With ↑sing atomic no., the atomic and ionic radii decreases from one element to the other but the decrease is very small.

★ A steady decrease in the size of lanthanoids with ↑se in atomic no. is known as Lanthanoid Contraction.

Consequences of Lanthanoid Contraction

a) It leads to similar physical & chemical properties among lanthanoids.

b) Zr and Hf have similar properties due to similar atomic radii .

c) Chemical separation of lanthanoids becomes difficult

Properties of Lanthanoids

1. Mainly give +3 O.S. but some elements show +2 and +4 O.S.
2. Silvery white soft metals & tarnish rapidly.
3. Hardness ↑ as atomic no. ↑ [Samarium (Sm) being steel hard]
4. Malleable, ductile, High M.Pt, good conductor of electricity, Highly dense metals.
5. Form alloys with other metals easily e.g. Mischmetall (consists of a Lanthanoid metal [~95%] and iron [~5%] and traces of S, C, Ca and Al)
6. Colour Many trivalent lanthanoid ions (Ln^{3+}) are coloured both in solid & in aq. solution. (Colour → due to the presence of f-electrons [f-f-transition])

→ $_{58}Ce$: $[Xe]\,4f^1 5d^1 6s^2$

Ce^{+4} : $[Xe]$

↳ attains Noble gas E.C

→ unpaired e's [f-f transition]

$f^0 - f^{14}$ → colourless

Homework

A) Electrochemistry (Ch-2) (75%)

B) Coordination Compounds

→ DPP Solve

→ Read NCERT + Q's.



THANK YOU